

# Toy Objects Foreground-Masking Optimisation

*Four-fold cross-validation for SEP and MTOjects*

Dataset: 40 low-foreground S4G galaxies | Application set: 182 galaxies

## 1. Purpose and scope

This document defines the reproducible optimisation design used to calibrate foreground-object masking with injected Toy Objects. SEP is optimised first; the complete procedure is then repeated for MTOjects. The design separates parameter fitting from held-out evaluation and retains a common independent injection set for comparing the four candidate solutions.

The workflow produces:

- four optimisation runs per algorithm, each trained on 30 galaxies;
- held-out evaluation on the remaining 10 galaxies, rotating once through every galaxy;
- cross-evaluation of all four candidate parameter sets on a common 40-galaxy Toy Object realization;
- a selected best-parameter JSON compatible with the canonical 182-galaxy batch tools; and
- CSV/workbook evidence supporting parameter and performance review.

## 2. Input sample validation

CleanGalaxies.txt is treated as the authoritative list. Before optimisation, the driver requires exactly 40 nonblank, unique names. Every name must occur in the 182-row geometry manifest, have all required bar and disc geometry fields, and resolve to an accessible 3.6-micron FITS image. Validation failure stops the run.

| Validation | Acceptance criterion                                                      |
|------------|---------------------------------------------------------------------------|
| Count      | Exactly 40 unique galaxy identifiers                                      |
| Manifest   | Every identifier matches one manifest row                                 |
| Geometry   | Centre, position angles, inclination, bar size and pixel scale are finite |
| Image      | The local or manifest FITS path exists                                    |

### 3. Toy Object construction

Six synthetic objects are injected into each science image inside the investigated, bar-aligned region. Objects are sampled as stars, compact clusters and elliptical galaxies. Their peak brightness is scaled to the robust image noise, making the task comparable across galaxies with different backgrounds.

| Property          | Sampling rule                                                      |
|-------------------|--------------------------------------------------------------------|
| Object type       | Star 50%; cluster 20%; galaxy 30%                                  |
| Peak              | Uniform from 5 to 25 robust-sigma                                  |
| Star/cluster FWHM | Uniform from 2 to 10 pixels                                        |
| Galaxy FWHM       | Uniform from 5 to 22 pixels                                        |
| Galaxy axis ratio | Uniform from 0.35 to 0.95                                          |
| Truth footprint   | Model pixels at least 8% of peak, then one-pixel circular dilation |

### 4. Cross-validation design

A fixed random seed shuffles the 40 names and distributes them into four disjoint folds of 10. For fold  $k$ , the other 30 galaxies form the training set and fold  $k$  is held out. Thus every galaxy is excluded from optimisation exactly once and used for held-out validation exactly once.

| Stage               | Images | Role                                                            |
|---------------------|--------|-----------------------------------------------------------------|
| Fold optimisation   | 30     | Choose parameters using the Toy Objects objective               |
| Held-out validation | 10     | Measure generalisation without influencing Optuna               |
| Common evaluation   | 40     | Compare all four candidates on identical independent injections |
| Final application   | 182    | Generate the canonical per-galaxy diagnostic outputs            |

Each fold runs 40 Optuna trials: eight broad startup samples followed by 32 Tree-structured Parzen Estimator trials. Ten image workers evaluate galaxies concurrently within a trial; trials themselves remain sequential so the adaptive sampler sees every completed result.

## 5. Objective function

The injected-image mask is compared with the baseline mask from the same unmodified galaxy. Only incremental mask pixels are scored. This isolates the response to the known Toy Objects from pre-existing detections.

incremental mask = injected-image mask AND NOT baseline mask

Per-image measurements include truth-pixel recall, precision, F-score, mean per-toy recall, the proportion of toys with at least 50% recall, incremental masked fraction, false-positive fraction and segment count.

### 5.1 SEP score

recovery = 0.45 mean recall + 0.20 mean F-score + 0.25 mean toy recall + 0.20 toy detection rate

score = recovery - 0.35 mean masked fraction - 0.05 false-positive fraction

### 5.2 MTOjects score

recovery = 0.45 mean F-score + 0.35 mean toy recall + 0.20 toy detection rate

MTOjects recovery = 0.45 mean F-score + 0.35 mean toy recall + 0.20 toy detection rate; loss = 0.50 mean masked fraction + 0.10 false-positive fraction; feasible score = recovery - loss.

For both algorithms, any trial whose worst individual image masks more than 15% receives a large cap-excess penalty. Optuna minimises objective = -score for feasible trials. MTOjects additionally rejects an empty incremental mask, toy detection below 25%, or mean toy recall below 20%. A candidate is released only when its common-40 detection is at least 50%, common-40 mean toy recall is at least 30%, and every held-out fold has non-zero recovery.

## 6. SEP parameter constraints

SEP Toy Objects detection is constrained to the original science image; residual-image detection is not permitted in the current optimiser or production batch. Parameter ranges follow the Source Extractor guide supplied with the project: relative thresholds approximately 0.6-2 sigma, minimum areas commonly 5-35 pixels, 32 deblend levels as a standard starting point, deblend contrast of order 0.01, and background meshes larger than the objects of interest. Project-specific post-detection filters retain their established ranges.

| Parameter       | Optimised range/choices         | Purpose                       |
|-----------------|---------------------------------|-------------------------------|
| detect_thresh   | 0.6-2.0 sigma                   | Relative detection threshold  |
| minarea         | 5-35 pixels                     | Minimum connected area        |
| deblend_nthresh | 16, 32 or 64                    | Deblending levels             |
| deblend_cont    | 0.001-0.03, log scale           | Minimum branch contrast       |
| back_size       | 32, 48, 64, 96, 128, 192 or 256 | Background mesh size          |
| filter_size     | 1, 3, 5, 7 or 9                 | Background filter size        |
| dilation_radius | 1-6 pixels                      | Final mask expansion          |
| max_area        | 20-8000 pixels                  | Reject overlarge segments     |
| max_elongation  | 1.5-30                          | Reject extreme segment shapes |

## 7. MTOjects parameter constraints

MTOjects uses the parameter domains established by the existing project optimiser and interactive model. Algorithm defaults not listed below remain fixed, including alpha, soft bias, gain and background mean.

| Parameter     | Optimised range           | Purpose                                  |
|---------------|---------------------------|------------------------------------------|
| move_factor   | 0.0-1.0                   | Tree-node movement control               |
| min_distance  | 0.0-1.0                   | Minimum hierarchy distance               |
| gaussian_fwhm | 0.0-5.0 pixels            | Optional Gaussian smoothing              |
| bg_variance   | 0.0001-10000, step 0.0001 | Background variance supplied to MTOjects |
| minarea       | 1-80 pixels               | Minimum accepted segment area            |

| Parameter       | Optimised range | Purpose                       |
|-----------------|-----------------|-------------------------------|
| dilation_radius | 0-8 pixels      | Final mask expansion          |
| max_area        | 20-3000 pixels  | Reject overlarge segments     |
| max_elongation  | 1.5-20          | Reject extreme segment shapes |

## 8. Candidate selection

The best training solution from each fold is first reported against its own held-out 10 galaxies. Because the four held-out sets contain different galaxies, those four scores are diagnostic rather than directly interchangeable. For a fair final comparison, every candidate is therefore scored on the same 40 galaxies using a separate, fixed Toy Object injection seed. The candidate with the smallest all-40 objective is selected.

This common evaluation still uses the same underlying science galaxies, so it is an independent injection test rather than a completely external astronomical sample. That limitation should be retained when interpreting results.

## 9. Recorded outputs and review

| Output                                     | Contents                                                        |
|--------------------------------------------|-----------------------------------------------------------------|
| cross_validation_config.json               | Seeds, folds, constraints and run settings                      |
| training_names.txt /<br>held_out_names.txt | Exact membership for each fold                                  |
| fold optimisation summary CSV              | All 40 trials and parameter combinations                        |
| cross_validation_candidates.csv            | Training, held-out and common all-40 metrics for each candidate |
| held_out_details.csv                       | Per-galaxy held-out performance                                 |
| cross-validation best JSON                 | Selected parameters and provenance for the 182-galaxy batch     |
| results workbook                           | Review-oriented sheets for assignments, candidates, details and |

| Output | Contents        |
|--------|-----------------|
|        | trial histories |

Review should focus on both performance and stability: the spread of parameters across folds, held-out score variation, false-positive masking, worst-image masked fraction, and whether Optuna repeatedly selects a bound. A high recovery score is not acceptable if it is achieved through excessive removal of galaxy data.

## 10. Final 182-galaxy processing

After candidate selection, the canonical batch runner applies the winning parameters to all 182 manifest galaxies. Each algorithm writes per-galaxy diagnostic PNGs and a summary CSV with status, segment counts, masked fraction, runtime and errors. Each method produces one aligned eight-panel PNG per galaxy: top row original and original plus toys; second row mask and recovered image with green correct and red incorrect outlines; third row original and processed isophotes; fourth row original and processed bar-major profiles. The processed profile retains the log-linear bridge: dashed green segments mark masked toy objects and dotted red segments mark all other masked items. The profile x-axis limits and physical panel widths match the image panels above. Calibration-galaxy filenames receive the `\_clean` suffix automatically.

## 11. Reproducibility and restart behaviour

All random seeds and fold assignments are saved. Optuna studies use SQLite storage, and the cross-validation drivers detect completed fold summaries before rerunning. A stopped workflow can therefore reuse completed folds. Terminal logs record progress, rough ETA and expected completion. Output folders are timestamped to prevent one study from overwriting another.

## Systematic documentation review - 17 August 2026

**Review classification:** Updated - current objectives, selection gates and eight-panel output

**Updates made:** The MTOBJECTS recovery feasibility penalty and final release gate were added, and the obsolete six-panel statement was replaced by the required aligned eight-panel report specification. Final selected folds were recorded.

**Current interpretation:** The design is complete and statistically defensible: four rotations of 30 training and 10 held out, followed by common-40 candidate comparison. SEP fold 4 and MTOBJECTS fold 3 are the selected solutions.

Authoritative combined outcome: SEP and MTOjects Masking Comparison and Recommendations  
20260817.docx. Full optimisation history and final science-image SEP fold records: Foreground Masking  
Optimisation Results.xlsx.